AquaSentinel

Finding water leaks underground, using only the sensors a utility already has

17/19	0	14 days	37%
LEAKS DETECTED (held-out year)	FALSE ALARMS in a full year	MEDIAN TIME to detection	CORRECT DISTRICT first guess (8% random)

What this document is. A complete, plain-language explanation of the project: the real-world problem, where the data came from, how the method works, what the results are, and where it falls short. It assumes no prior knowledge of water engineering. Section 6 walks through one real leak from start to finish - that worked example is the fastest way to understand everything.

Dataset: BattLeDIM 2020 (Zenodo record 4017659, CC BY 4.0), released by the KIOS Center of Excellence, University of Cyprus.
Network: L-Town - 782 junctions, 905 pipes, 43 km, 33 pressure sensors.
Protocol: 2018 used for all development; 2019 held out entirely and evaluated once.

Everything in this document is regenerated directly from the result tables produced by the code, so the figures quoted here and the output of the software cannot drift apart. Generated 16 October 2026.

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1. The project in one page

Cities lose an enormous amount of treated water through leaks in buried pipes that nobody knows about. A burst that breaks the road surface gets reported within hours. A small crack simply pushes water into the soil, silently, for months. This project builds a system that spots those hidden leaks early and tells a repair crew which district to search.

It produces two outputs:

- **Detection** - an alarm saying a *new* leak has started, and roughly when.
- **Localisation** - which of 12 districts to search, narrowing the hunt from 43 km of pipe to about 3.6 km.

The inputs are only what a water utility already measures: 33 pressure sensors, 3 flow meters and 1 tank level sensor, sampled every five minutes. No new hardware, no fieldwork.

Headline results

Evaluated against the method utilities use today, **minimum night flow** (MNF), on a year that played no part in developing the system:

Measure	This project	Night flow (current practice)
Leaks detected (2019)	**17 of 19 (89%)**	1 of 19 (5%)
False alarms	**0**	0
Median time to detection	**14 days**	30 days
Leaks detected (2018)	13 of 14 (93%)	1 of 14 (7%)

> **The one idea to take away.** A leak makes water rush toward the hole, which increases friction and *drops the pressure* nearby. Everything in this project is a careful way of measuring that drop and working out where it came from.

2. The physical setting

What a water distribution network is

A city's water supply is a network of buried pipes - much like a road network, but for water. Water enters at a few points (reservoirs, a pump, a storage tank) and travels through the pipes to every building.

The network in this study is called **L-Town**:

- **782 junctions** - points where pipes meet and where buildings draw water
- **905 pipes** - about **43 km** of pipe in total
- **2 reservoirs, 1 pump, 1 storage tank** - the sources of supply
- **3 pressure reducing valves (PRVs)** - valves that cap the pressure

What pressure is, and why it matters

Pressure is how hard water pushes inside a pipe. Here it is measured in **metres**: '50 m of pressure' means the water pushes as hard as a 50-metre column of water would. Enough pressure is needed to reach upper floors; too little and taps run dry.

What a leak does physically

A leak is a hole. Water escapes through it, and that escaping water has to come from somewhere - so extra water is pulled through the surrounding pipes toward the hole. When more water moves through a pipe, friction rises and **pressure falls**. The drop is largest near the leak and fades with distance.

> **This is the entire physical basis of the project.** A leak creates a pressure drop whose *size* tells you how big the leak is, and whose *pattern across sensors* tells you where it is.

The instruments available

The utility has only this much visibility into 43 km of buried pipe:

- **33 pressure sensors** at 33 junctions, reporting every 5 minutes
- **3 flow meters** measuring water entering the network
- **1 tank level sensor**

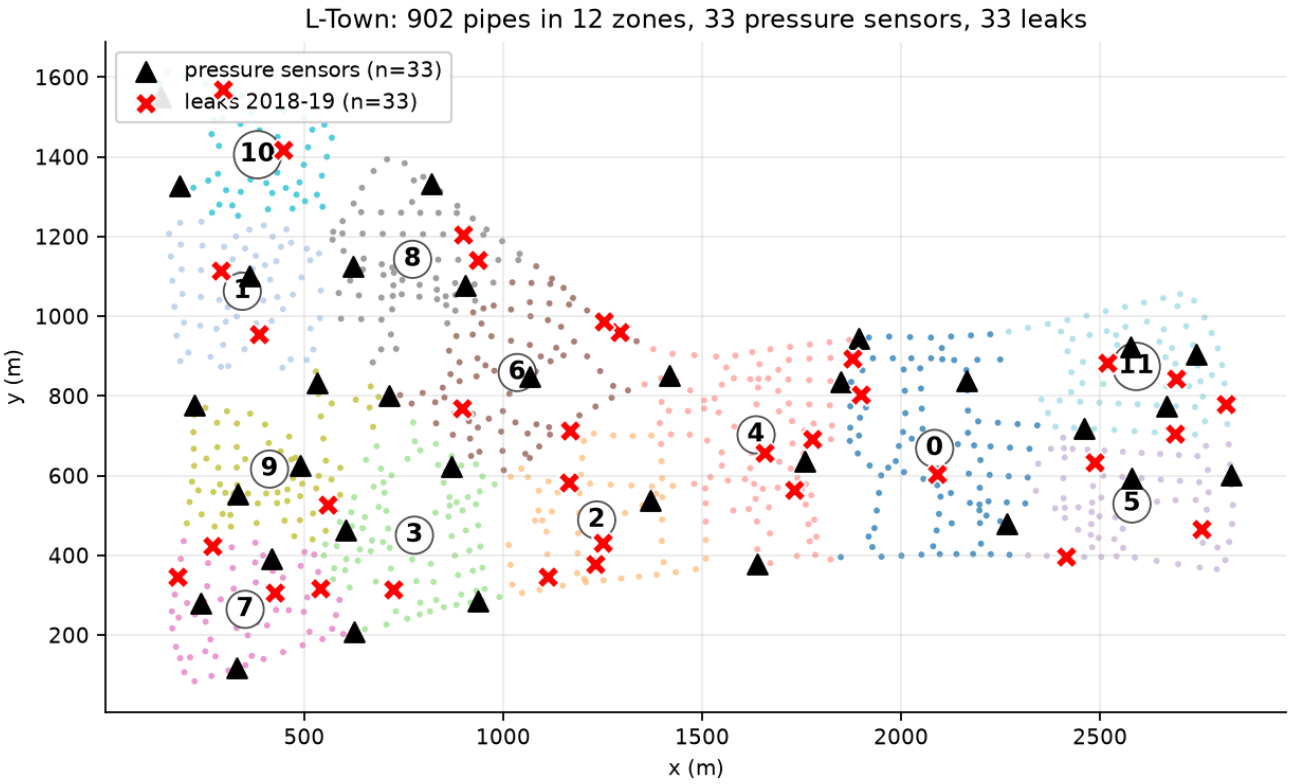


Figure 1. The L-Town network. Coloured dots are pipes, shaded by district. Black triangles are the 33 pressure sensors - the only inputs the system gets. Red crosses mark the 33 real leaks. Note how sparse the sensors are relative to the pipes: that sparsity is the core difficulty of the problem.

3. The real-world problem

Water lost between treatment and the customer's meter is called **non-revenue water** - water that was cleaned, pumped and paid for, but never billed. Most of it is leakage. The cost is not only the water itself but the chemicals and electricity spent producing and moving it, plus the damage done underground before anyone notices.

What utilities do today: minimum night flow

Between roughly 2 am and 4 am almost nobody uses water. So whatever is still flowing into a district at 3 am is assumed to be loss. If that night-time flow steps up, a new leak is suspected. It is simple, it needs no modelling, and it is the standard method in the industry.

Why it performs poorly here

Night flow is noisy. Some nights people are awake; some nights are warmer. Measured on this network:

Quantity	Magnitude
Random night-to-night variation	26 to 52 m³/h
Extra flow caused by a typical leak	10 to 45 m³/h

The noise is larger than the signal. The leak is buried inside ordinary variation, so it simply cannot be seen. Implemented faithfully as a baseline, this method found **1 leak out of 19**.

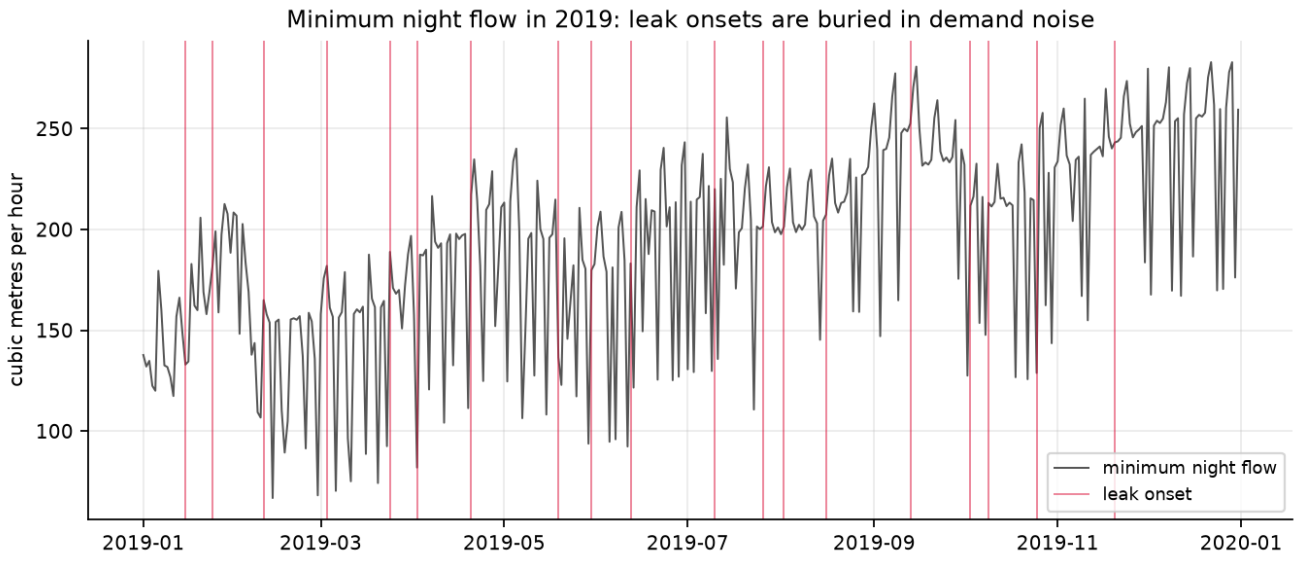


Figure 2. Night-time inflow through 2019. Each red line is a real leak starting. The onsets are indistinguishable from ordinary fluctuation - which is precisely why a better method is needed.

4. The data: where it comes from and what is in it

Source

- **Dataset:** BattLeDIM 2020 - 'Battle of the Leakage Detection and Isolation Methods'
- **Produced by:** the KIOS Center of Excellence, University of Cyprus
- **Why it exists:** it was released for an international research competition in 2020, so that competing methods could be compared on identical data with identical answers
- **Obtained from:** Zenodo, a public research repository - record 4017659, licence CC BY 4.0 (free to use with attribution)
- **How:** the script `src/download_data.py` fetches it automatically and resumes if the connection drops

> **Is it real data?** It is generated by hydraulic simulation of a real network model, with realistic demand patterns and measurement noise, and was published as a benchmark precisely so that methods could be compared against *verified* ground truth. That is a strength rather than a weakness: real utility records almost never come with confirmed leak labels, so honest measurement of detection rate and timing would be impossible.

What the files contain

File	Contents
L-TOWN.inp	The network model: every pipe, junction, diameter and elevation
2018_SCADA_Pressures.csv	33 pressure sensors, every 5 minutes, all year
2018_SCADA_Flows.csv	3 flow meters
2018_SCADA_Levels.csv	Tank level
2018_Leakages.csv	*Answer key* - flow through each leak at every moment
dataset_configuration.yaml	*Answer key* - pipe, start, end, hole diameter and type of every leak
(the same files for 2019)	The held-out evaluation year

SCADA stands for Supervisory Control and Data Acquisition - the industry term for the system that gathers sensor readings.

What a row of data looks like

Timestamp	n1	n4	n31	n54	n105	n114
2019-01-01 00:00:00	28.63	33.72	37.00	36.93	50.45	53.92
2019-01-01 00:05:00	28.66	33.75	37.02	37.05	50.54	54.02
2019-01-01 00:10:00	28.67	33.76	37.04	37.03	50.51	53.98

Each row is one instant; each column is one sensor's pressure in metres. There are **105,120 rows per year** (365 days × 24 hours × 12 readings per hour).

The ground truth

Every leak is documented exactly. For example:

> `p523,  2019-01-15 23:00,  2019-02-01 09:50,  0.020246 m,  abrupt`
>
> Read as: *pipe p523 leaked from 15 January at 11 pm until 1 February, through a 20.2 mm hole, and it was an abrupt burst.*

There are **33 such leaks** - 14 beginning in 2018 and 19 in 2019.

Two kinds of leak - this drives every result

Type	What it is	Behaviour	Count
Abrupt	A burst - the pipe cracks open at once	Full discharge from the first second. Comparatively easy to spot.	15
Incipient	A crack that slowly widens over weeks	Starts almost invisibly and grows. Much harder to catch early.	18

5. The discovery that shaped the whole project

The obvious first approach is a classifier: feed in the sensor readings, and predict 'leak' or 'no leak'. Checking the labels rules that out immediately.

Year	Share of the year with no leak flowing	Most leaks running at once
2018	2.2%	6
2019	**0%**	**16**

In 2019, **something is leaking at every single moment of the year**. A model that always answers 'yes, there is a leak' would be 100% accurate and completely worthless - it tells an operator nothing they can act on.

There is a second difficulty. To recognise abnormal behaviour you need examples of normal behaviour, and there are only eight leak-free days in the entire dataset - all in January. A model trained on those would know what normal looks like *in winter*, and would then have to extrapolate across a whole year of changing demand.

So the question was changed.

Rejected: *'Is there a leak right now?'* - unanswerable on this data, and useless even if answered.
Adopted: *'Has a **new** leak just started?'* - answerable, and exactly what a control room needs.

This reframing is the project's main conceptual contribution. Every design decision that follows exists to serve it.

6. How the system works, step by step

The five steps below are traced through one real leak, so that each stage can be seen doing its job on actual data.

> **The worked example.** Pipe **p523** burst on **15 January 2019 at 11 pm**, through a 20.2 mm hole. It is an *abrupt* leak. Follow this one case through all five steps and the whole method becomes concrete.

Step 1 - Predict what each sensor should be reading

For each of the 33 pressure sensors, a small statistical model (**Ridge regression**, a standard and very stable linear method) predicts that sensor's pressure from four things:

- total inflow entering the network
- inflow squared, because friction rises faster than flow does
- the storage tank level
- time of day and day of week

Those inputs matter because pressure changes constantly for perfectly innocent reasons. At 8 am everybody showers, demand spikes and pressure falls across the whole network. That is not a leak. By giving the model the time of day and the inflow, it learns the normal daily and weekly rhythm - so the morning peak is never mistaken for a burst.

Step 2 - Compare against the last three weeks, not against perfection

This is the design choice that makes the method work despite there being no clean data. Rather than training once on some ideal healthy period, the model is **refitted every day on the previous 21 days**.

Three consequences follow, and all three are essential:

- **Existing leaks go quiet.** A leak that has been running for a month sits inside the 21-day reference, so the model already expects it. It cannot keep raising alarms.
- **Seasons take care of themselves.** The reference is never more than three weeks old, so summer is always compared against summer.
- **Only new leaks stand out.** Something that began yesterday is not yet in the reference, so it breaks the learned relationship.

A refinement: the two most recent days are excluded from the reference window, so a leak that started yesterday cannot slip into its own baseline and hide itself.

Step 3 - The residual: the signal everything rests on

Residual = what the sensor actually read – **what the model predicted.**

When the network behaves normally the residual sits near zero. When a leak starts, real pressure falls below the prediction and the residual turns negative. Here is the real recorded residual at sensor **n506** around the p523 burst:

Date	Residual (metres)	Interpretation
11 Jan	-0.001	normal
12 Jan	+0.006	normal
13 Jan	+0.021	normal
14 Jan	-0.004	normal
15 Jan	-0.008	normal - the leak begins at 11 pm tonight
16 Jan	**-0.311**	**pressure is 31 cm below prediction**
17 Jan	-0.302	still there
18 Jan	-0.305	still there
19 Jan	-0.280	still there
20 Jan	-0.248	still there

Before the leak the model predicts the sensor to within about **2 centimetres**. The moment the leak starts the error jumps to **31 centimetres** - roughly **15 times larger** - and stays there. Detecting that jump, reliably and without crying wolf, is what the rest of the system does.

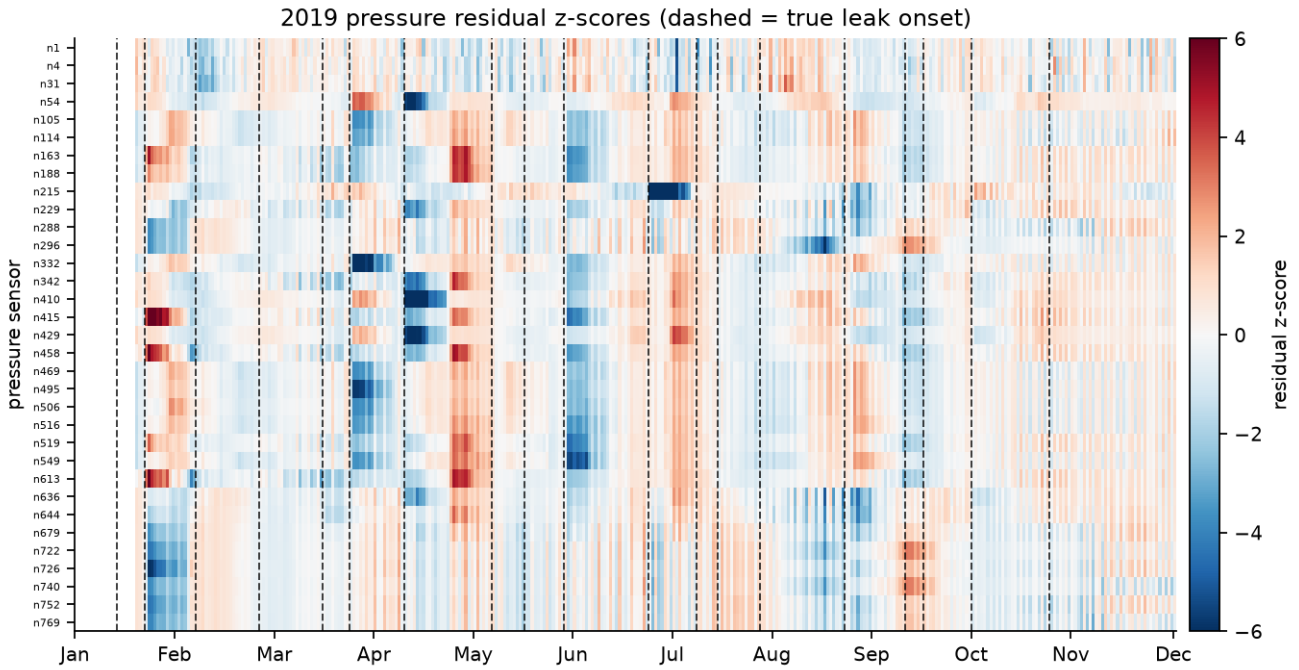


Figure 3. Residuals for all 33 sensors across 2019, expressed as z-scores. Each row is a sensor, each column a day. Dashed lines mark true leak onsets. Coloured bands appearing and persisting after a dashed line are leaks making themselves visible.

Step 4 - CUSUM: deciding when to raise the alarm

A simple rule such as 'alarm if the residual is large' would work for bursts but fail for slow cracks, which never produce a single dramatic day - only a long run of mildly odd ones. So the system uses **CUSUM** (cumulative sum).

> **CUSUM in one analogy: a bank balance.** Every abnormal day pays money in. Every normal day takes money out. When the balance crosses a set level, the alarm fires. That is why it catches slow leaks: no single day is convincing, but ten mildly abnormal days in the same direction add up to proof.

After an alarm the balance resets and the system stays silent for 14 days - otherwise a single unrepaired leak would alarm every day for a month.

For p523: evidence accumulated until the threshold was crossed on **25 January** - an alarm **10 days** after the burst began.

How the threshold was chosen: many settings were tested *on 2018 only*, keeping the one that caught the most leaks while raising no more than six false alarms a year. It was then frozen and never revisited. The false-alarm budget - not raw accuracy - is the binding constraint, because a system that cries wolf gets switched off by the people who have to respond to it.

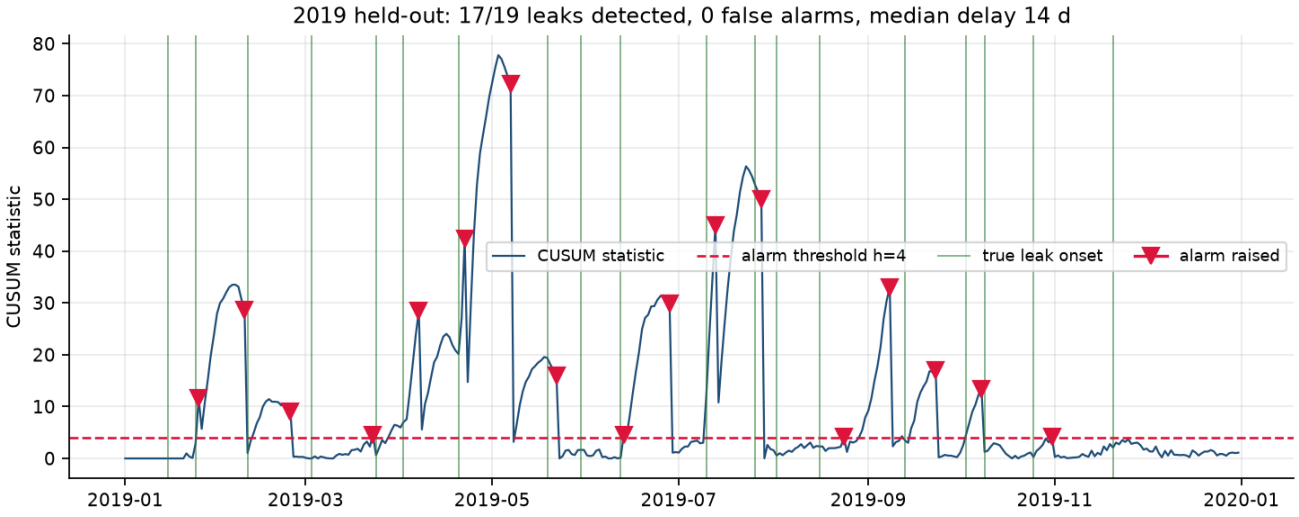


Figure 4. The detector running across the held-out year. The blue line is accumulated evidence, the dashed red line the alarm threshold, green lines the true leak onsets and red triangles the alarms raised. Every alarm follows a genuine onset, and no alarm appears anywhere else.

Step 5 - Localisation: working out where to dig

Detection says something broke. Localisation says where to send the crew.

The idea - a pressure fingerprint. When a leak starts, all 33 sensors shift, but by different amounts: nearby sensors drop sharply, distant ones barely move. Collecting those 33 shifts into a list gives a fingerprint with two separable properties - its *shape* (which sensors moved most) indicates **where** the leak is, while its overall *size* indicates **how big** it is. Since leak size is unknown, the size is normalised away and only shape is compared.

The obstacle. Learning 'this fingerprint means district 7' needs many examples per district. There are only 14 training leaks across 12 districts - roughly one each. Far too few to learn from.

The solution - generate the examples from physics instead. Using **EPANET**, the free water-network simulator published by the US Environmental Protection Agency and the worldwide standard for this purpose (driven from Python through the **WNTR** library), a leak is placed on each of the **902 pipes** in turn, the hydraulics are solved, and the resulting 33-sensor response is recorded. The result is a reference library of 902 fingerprints derived from physics rather than from data.

The simulated hole is set to 15 mm - the median of the real leaks - giving about 14.2 m³/h of discharge. The resulting average pressure response of 0.29 m sits squarely inside the range of the real observed shifts (0.03-0.48 m), which is good evidence the simulation is realistic.

An observed fingerprint is then compared against all 902 simulated ones using **cosine similarity** - a standard measure of whether two lists of numbers have the same shape, where 1 means identical and 0 means unrelated. The best-matching pipes lend their scores to their districts.

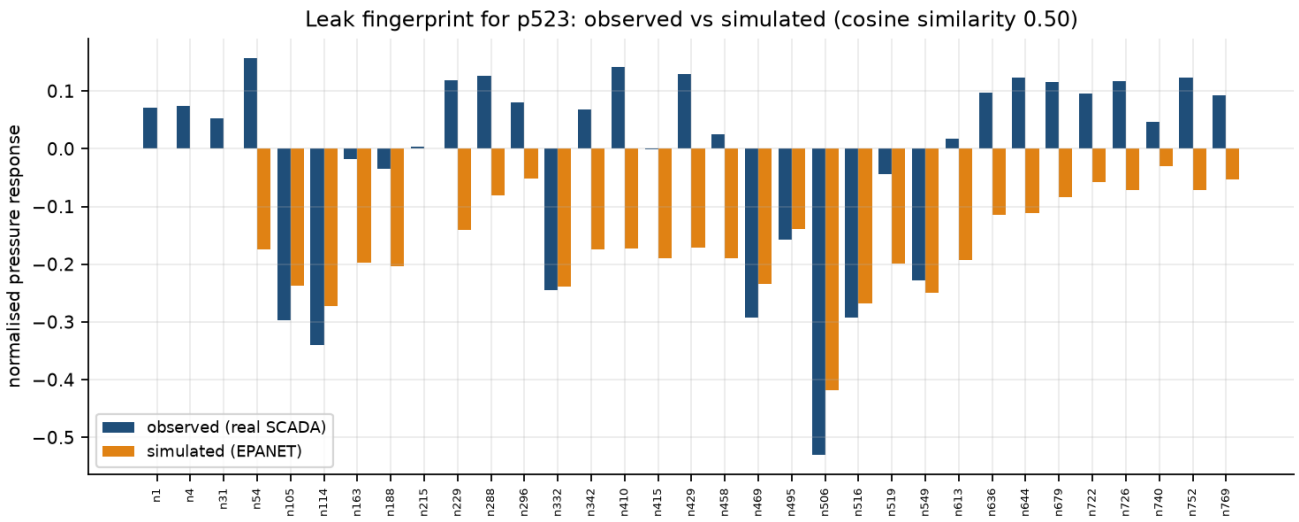


Figure 5. The fingerprint of leak p523: what the real sensors did (blue) against what EPANET predicted for that pipe (orange). The two rise and fall together. That agreement between independent measurement and physical simulation is what makes localisation possible.

Why districts rather than individual pipes

With 33 sensors and 905 pipes, two neighbouring pipes simply cannot be told apart - there is not enough information in the measurements. This is a mathematical limit, not a shortcoming of the code, and no method can escape it. The network is therefore divided into **12 districts** of roughly 3.6 km of pipe each (using k-means clustering on pipe positions), which is also the granularity at which a repair crew is actually dispatched.

The result for p523

Predicted district	True district	Rank of correct district	Best-matching pipe	Distance from the real leak
7	**7**	1 of 12	p498	**53.8 m**

The whole story in one sentence. The leak bursts on 15 January at 11 pm; next morning sensor n506 reads 31 cm below prediction; CUSUM accumulates evidence for ten days and alarms on 25 January; the fingerprint is matched against 902 simulated ones; the system says 'search district 7' - which is correct, and its single best guess is a pipe 54 metres from the actual hole.

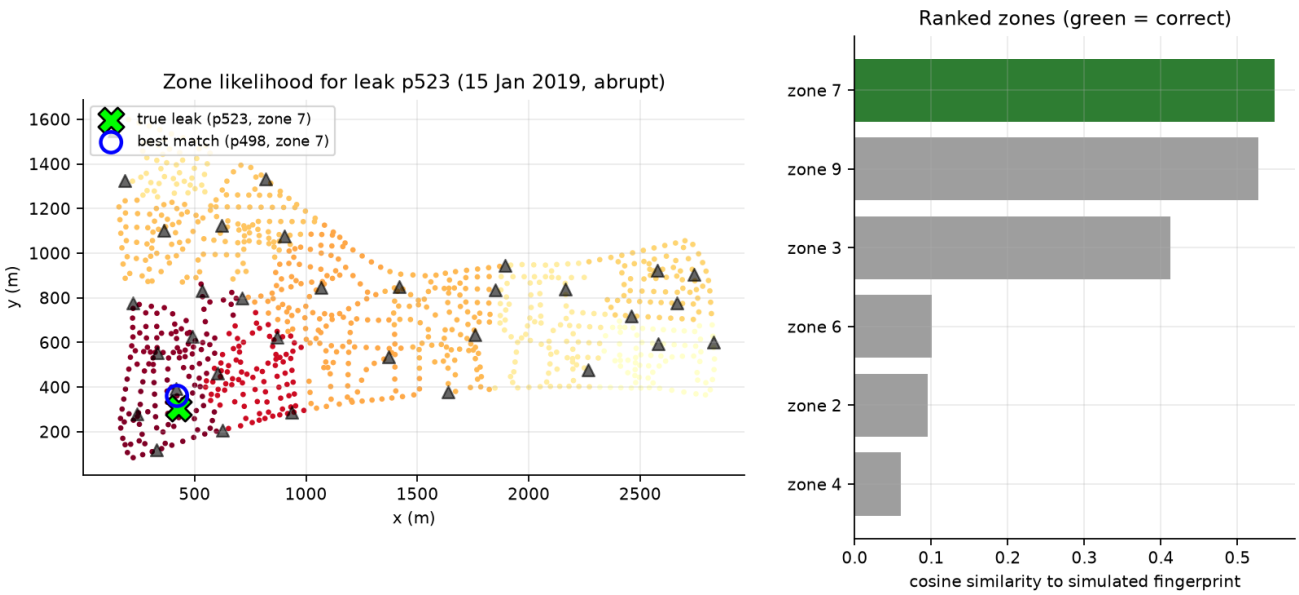


Figure 6. Leak p523 being localised. Left: every pipe shaded by how likely its district is, with the true leak marked in green and the best-matching pipe circled. Right: the districts ranked by score, the correct one shown in green.

7. Results

The testing rule that makes these numbers trustworthy

> **2018** was the practice year - every threshold, window length and design choice was selected using it.
> **2019** was the examination - run *once*, at the end, with everything frozen, and whatever came out is reported here.
>
> This is called a **held-out test set**. Without it, good-looking results mean very little, because any method can be tuned until it fits the data it was tuned on.

Detection

Year	Role	Leaks	Detected	False alarms	Median delay	Night-flow baseline
2018	development	14	**13 (93%)**	4	21 d	1 (7%)
2019	held-out	19	**17 (89%)**	0	14 d	1 (5%)

On the held-out year the system found **17 of 19 leaks**, raised **no false alarms at all**, and found them in a median of **14 days** - against 30 days and a single detection for the method in standard industrial use.

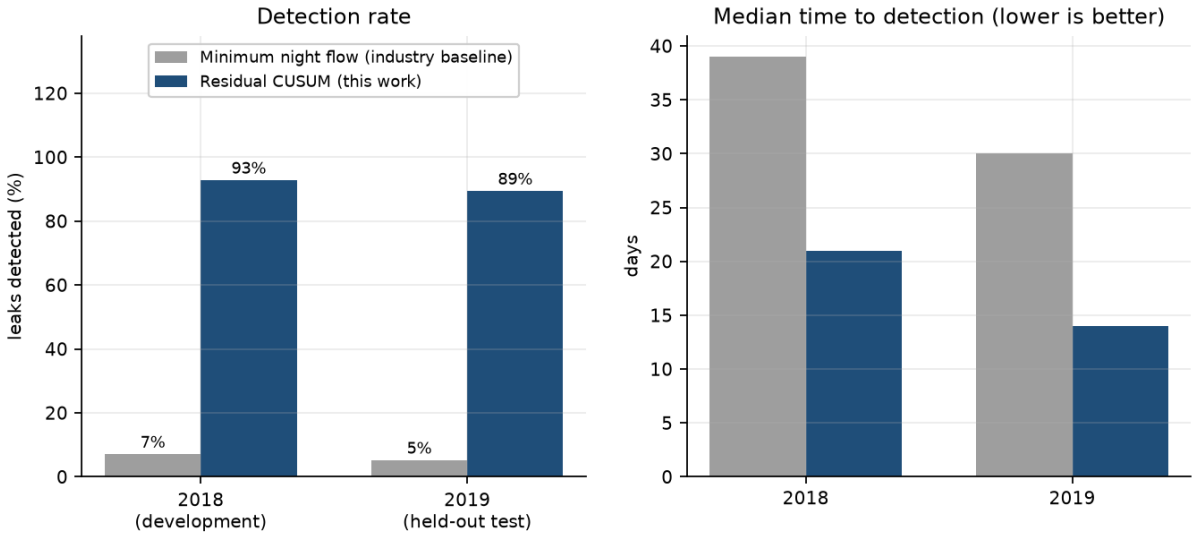


Figure 7. This method against minimum night flow, on both years.

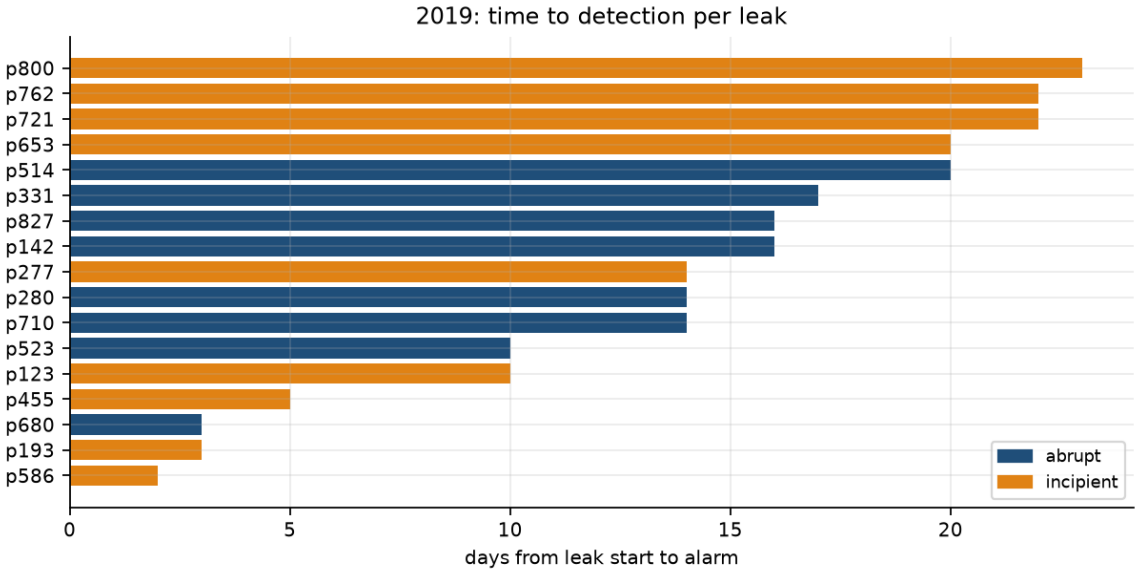


Figure 8. Time from leak start to alarm, leak by leak. Abrupt bursts (blue) are generally caught faster than slow incipient cracks (orange).

Localisation

Year	Leaks	Correct district (top-1)	Within top 3	Median rank	Median distance error
2018	12	**58%**	58%	1 of 12	324 m
2019	19	**37%**	53%	3 of 12	518 m
random guessing	-	*8%*	*25%*	*6.5*	-

Broken down by leak type, the difference between bursts and slow cracks is consistent across both years:

Year	Leak type	Number	Top-1	Top-3
2018	abrupt	6	100%	100%
2018	incipient	6	17%	17%
2019	abrupt	9	44%	56%
2019	incipient	10	30%	50%

In operational terms: searching blind means covering **43 km** of pipe. Naming the correct district reduces that to **3.6 km**, and a three-district shortlist to about **11 km** - a reduction of roughly **75%** in the ground a crew has to cover.

How many sensors are actually needed?

Sensors	Detection rate	False alarms per year	Median delay	Localisation top-1
5	98%	1.8	15.8 d	18%
10	100%	2.6	12.2 d	20%
15	100%	3.6	10.2 d	27%
20	100%	3.2	12.6 d	28%
25	96%	1.6	13.6 d	31%
33	**89%**	**0.0**	**14.0 d**	**37%**

Read this table carefully - it looks backwards at first. Detection rate on its own can be gamed: with fewer sensors the tuning simply settles on a lower threshold, which catches more leaks *and* raises more false alarms. Looking at both columns together gives the real conclusion: **five sensors are enough to know that something is wrong; it is the full network of 33 that removes false alarms entirely and doubles the ability to say where.** For a utility deciding where to spend money, that is the useful finding.

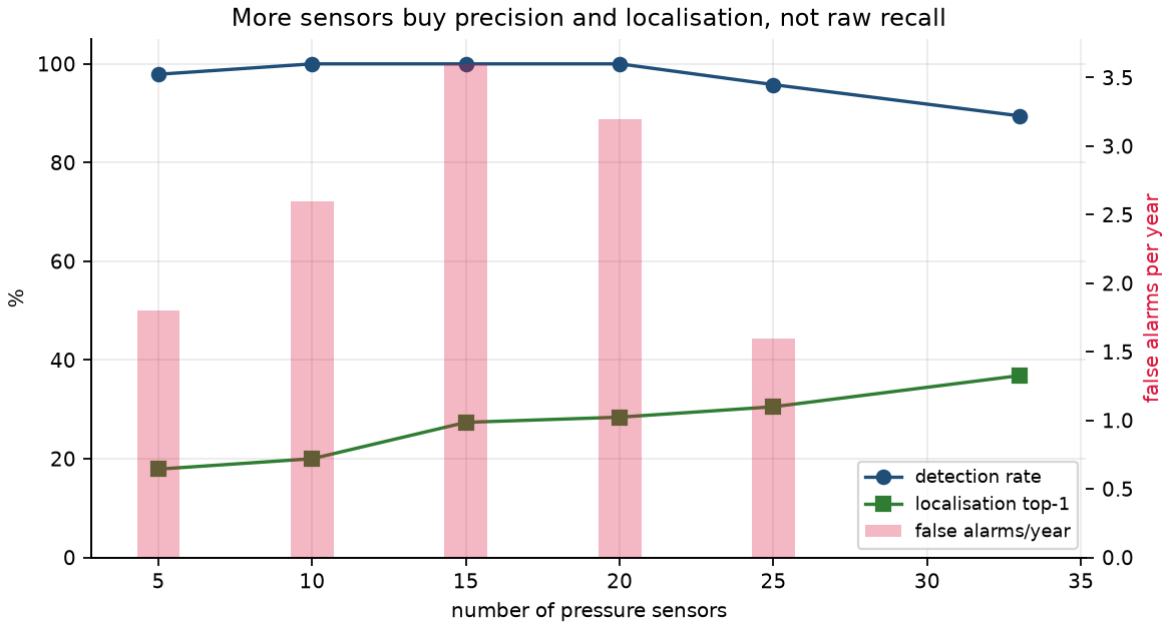


Figure 9. Detection, localisation and false alarms as the number of sensors varies. Sensors buy precision and location, not raw recall.

How finely can a leak be located?

Districts	Top-1	Top-3	Random top-1	Pipe per district
6	42%	84%	17%	7.2 km
8	42%	63%	12%	5.4 km
10	26%	58%	10%	4.3 km
12	37%	53%	8%	3.6 km
16	16%	53%	6%	2.7 km
20	37%	47%	5%	2.1 km

Performance relative to chance improves as districts get finer, but absolute accuracy falls and becomes unstable. With only 19 test leaks a single leak is worth five percentage points, so the jumps between 10, 12 and 20 districts are sampling noise rather than real structure. Twelve districts is a sensible operating point; the honest statement is that this quantity of data cannot pin down the optimum precisely.

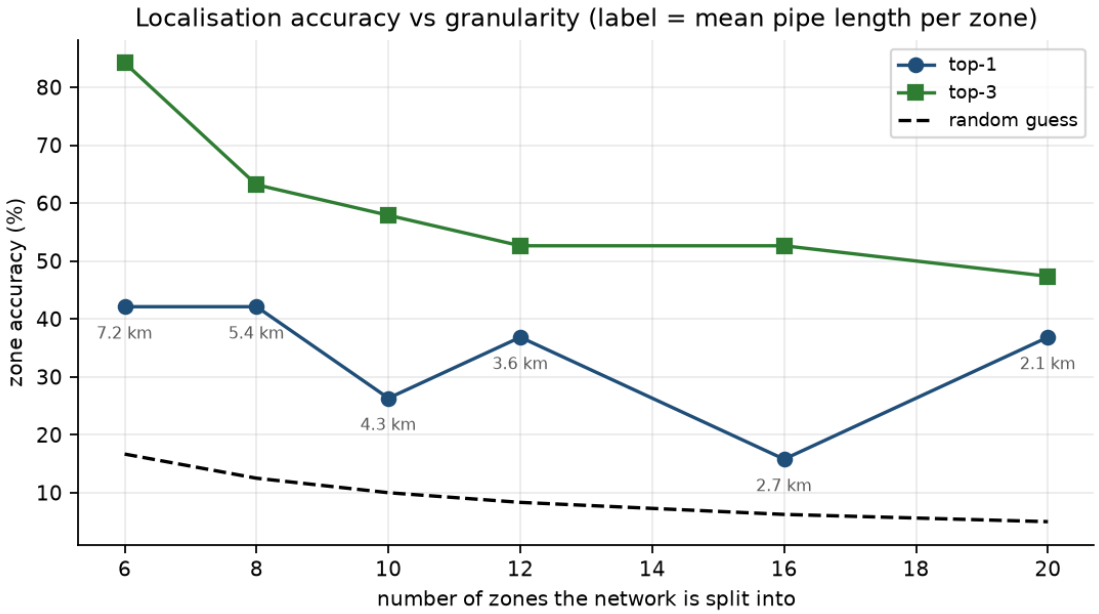


Figure 10. Accuracy against how finely the network is divided.

8. Limitations - stated plainly

Every method has a boundary. Naming these before anyone else does is part of presenting the work honestly.

- **Two leaks out of nineteen were missed** (p426, p879), both late in the year when eight other leaks were already running and masking further change.
- **Nineteen test leaks is a small sample.** One leak is worth about five percentage points, so no difference smaller than roughly ten points should be treated as meaningful.
- **Slow incipient leaks remain the weak case** for both speed of detection and accuracy of location - they are faint by nature in their early weeks.
- **Localisation is district-level, not pipe-level,** and with 33 sensors over 905 pipes that is a mathematical limit rather than an engineering one.
- **The simulated fingerprints carry model error.** The published network model is documented as differing from the true network by up to 10% in pipe diameters, roughness and demand, which caps match quality regardless of method.
- **Simulated leaks are attached to a pipe's upstream junction,** a small approximation that is acceptable at district granularity.
- **One network, two years.** Nothing here demonstrates that the method transfers to a different topology without re-tuning and a new simulated library.
- **Matching alarms to leaks is necessarily approximate.** With up to 16 leaks running at once, attributing an alarm to one specific onset is ambiguous; alarms are matched to the oldest unmatched leak within 45 days.

9. Running the code

On a machine that is already set up

Open a terminal in the project folder and run:

```
.venv/Scripts/python.exe run_all.py
```

Recomputes every result, table and figure. About 96 seconds using the cached data.

```
.venv/Scripts/streamlit run app.py
```

Opens the interactive dashboard in a browser. Ctrl+C in the terminal stops it.

From scratch on a new machine

```
python -m venv .venv
.venv/Scripts/python.exe -m pip install -r requirements.txt
.venv/Scripts/python.exe src/download_data.py
.venv/Scripts/python.exe run_all.py
```

The download is about 181 MB from Zenodo and resumes if interrupted. The first run_all takes roughly 15 minutes because it simulates a leak on all 902 pipes; afterwards everything is cached.

What each file does

File	Purpose
src/config.py	Paths, sensor lists and the ground-truth leak table
src/download_data.py	Fetches the dataset from Zenodo, resumable
src/data_loader.py	Reads the CSVs, builds labels and time features
src/network.py	Network topology, district clustering, geometry
src/residuals.py	The rolling-reference expected-pressure model
src/detect.py	CUSUM detector, threshold tuning, evaluation
src/simulate.py	EPANET simulation of a leak on every pipe
src/localize.py	Fingerprint matching to districts
src/experiments.py	Runs the whole study, writes reports/tables
src/figures.py	Produces every figure in this document
src/make_pdf.py	Builds this PDF
app.py	The interactive dashboard
run_all.py	Reproduces everything end to end

The dashboard

Four tabs:

- **Overview** - headline figures against current practice, a four-step summary of the method, and the pipe network drawn by district

- **Year replay** - a date slider that moves through the year: leaks appear when they start, turn green once the detector flags them, and alarms accumulate on the evidence timeline
- **Leak explorer** - choose any leak and see where the crew would be sent, with the network shaded by match strength, the districts ranked, candidate pipes and their distance from the real leak, and the fingerprint compared against simulation
- **Evidence & limits** - comparison charts, all result tables and the limitations

The sliders re-run the detector live, which is useful for showing how sensitive the results are to the thresholds. The reported results use the values tuned on 2018: **k = 1.0, h = 4, refractory period 14 days, 12 districts**. The sidebar confirms when those settings are active, and a **Reset to reported settings** button restores them.

10. Glossary

Term	Meaning
SCADA	Supervisory Control and Data Acquisition - the system that collects sensor readings from the network.
Residual	Actual sensor reading minus predicted reading. The core signal of this project.
Ridge regression	A simple, stable linear prediction model.
CUSUM	Cumulative sum - a running tally that accumulates evidence over days before raising an alarm.
z-score	How many standard deviations a value sits from normal; puts all sensors on a common scale.
MAD	Median Absolute Deviation - a robust measure of noise that a developing leak cannot inflate.
EPANET	The free water-network simulator published by the US EPA; the worldwide standard.
WNTR	Water Network Tool for Resilience - the Python library used to drive EPANET.
Emitter	How EPANET represents a leak: discharge depends on pressure, $Q = C\sqrt{P}$.
Cosine similarity	A measure of whether two lists of numbers have the same shape, independent of their size.
k-means	A clustering algorithm; used here to group pipes into geographic districts.
Minimum night flow (MNF)	The industry-standard leak indicator: flow still entering a district at 3 am.
Abrupt / incipient	A sudden burst / a crack that widens slowly over weeks.
Held-out test set	Data never used for any tuning decision, so the results on it are honest.
PRV	Pressure Reducing Valve - caps pressure in part of the network.
Non-revenue water	Water produced and paid for but never billed; mostly leakage.

11. Questions likely to be asked, and how to answer them

Why not use deep learning or a neural network?

The limiting resource is not data volume but *labelled events*. There are 210,000 timesteps but only 33 leaks, and just 14 available for training. A deep model would memorise those 14 examples. Ridge regression with a CUSUM is appropriate to the sample size, and CUSUM is in fact the statistically optimal detector for a persistent shift in mean - which is exactly what a leak is.

37% accuracy sounds low. Is that not a failure?

Against random guessing at 8% it is 4.4 times better, and the top-three figure of 53% compares against 25%. The more meaningful measure is operational: the search area drops from 43 km of pipe to 3.6 km. A crew that previously had no starting point now has a 37% chance of being in the right district immediately and 53% within three attempts.

Fourteen days to detect a leak seems slow.

It is half the 30 days taken by the method currently in industrial use. For incipient leaks part of the delay is physics rather than method: in the first days the crack is barely open and there is almost nothing to measure. Abrupt bursts are caught considerably faster.

How do you know you have not overfitted?

The 2019 data played no part in any decision. Every threshold, the signature window and the district count were chosen on 2018 and then frozen; 2019 was run once to produce the final numbers. Supporting evidence: the 2018 results are *better* (93% detection), which is exactly what one expects from the year used for tuning.

The dataset includes 82 household meters. Why were they not used?

Two reasons. Realism: in practice automated meter readings reach the utility with a long reporting delay, so they are not available for real-time detection. And discipline: the inputs were restricted to what a control room genuinely sees live - 33 pressures, 3 flows and one tank level.

Why match fingerprints instead of training a classifier for location?

With 14 real labelled leaks spread over 12 districts there is roughly one example per class, which cannot support a trained classifier. The simulated library supplies 902 reference fingerprints, but they are physics-derived rather than observed, so nearest-match comparison in signature space is the appropriate tool. It is the classical sensitivity-matrix approach from the hydraulics literature.

Why twelve districts?

Six through twenty were tested. Fewer districts are easier to hit but send the crew across more pipe - six districts gives 42% top-1 but each covers 7.2 km. Twelve is a reasonable operating point at 3.6 km each. With only 19 test leaks the data cannot resolve the true optimum, and the report says so.

What is genuinely new here?

Not any individual algorithm - Ridge regression, CUSUM and sensitivity matrices all exist. The contribution is the framing and the combination: recognising that 'is there a leak' is unanswerable on this data and reframing it to onset detection; using a rolling reference to cope with having essentially no leak-free data; and generating the localisation library hydraulically because the real one is far too small.

Would this work on a different network?

The method transfers, but it would need re-tuning and a new EPANET model to regenerate the fingerprint library. Transfer has not been demonstrated, and that is listed as a limitation.

What would be needed to deploy it for a real utility?

A calibrated hydraulic model of the network, a live feed from the SCADA historian, and a few months of data to establish the rolling reference. The computation is negligible - the entire study reruns in 96 seconds.

Two findings worth mentioning - they show real experimentation

The night-hours result. Restricting the analysis to night-time hours - the intuition on which night-flow monitoring is built - made performance *worse*, with signal-to-noise falling from 3.9 to 1.6. Discarding 80% of the day costs more in sample size than it gains in quietness. Once demand is modelled explicitly there is no reason to throw daytime data away.

The year-boundary result. An early version restarted the model each 1 January, leaving the detector blind for its first three weeks - exactly where two of the 2019 leaks begin. Allowing the reference window to reach back into the previous year, as a real utility's would, raised detection from 14 of 19 to **17 of 19** and halved median latency from 30 days to **14**.

The five sentences worth memorising

1. A leak causes a pressure drop, largest near the leak - that is the entire physical basis of the project.

2. 'Is there a leak' is useless here because something is leaking 100% of the time in 2019, so the question becomes 'has a *new* leak started'.

3. Each sensor's expected pressure is predicted from the previous 21 days, and a new leak shows up as that prediction suddenly failing - a 2 cm error becomes 31 cm.

4. With only 14 training leaks, the location fingerprints were generated by simulating a leak on all 902 pipes in EPANET rather than learned from data.

5. 89% of leaks found with zero false alarms, on a year never touched during development, against 5% for the method utilities use today.

References.

1. Vrachimis, S. G. et al. *Battle of the Leakage Detection and Isolation Methods.* Journal of Water Resources Planning and Management, 2022.
2. BattLeDIM 2020 dataset, Zenodo record 4017659 (CC BY 4.0), KIOS Center of Excellence.
3. Klise, K. A. et al. *WNTR: Water Network Tool for Resilience.* US EPA.
4. Rossman, L. A. *EPANET 2 Users Manual.* US EPA, 2000.